TFPT — Appendix H: the horizon unit system

One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

June 12, 2026

What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant $c_3 = \frac{1}{8\pi}$. This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound, $v_{\mathrm{GW}} = c$ and cosmic birefringence — in seam units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$, $|\mu_4| = 4$).

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus this appendix**, best read in order:

1. `introduction` — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

H. `tfpt_horizon_readouts` — **Appendix H**: one seam constant as the universal horizon thermal code (a unit-system reframe, *not* a peer paper, *not* new physics).

You are reading Appendix H — a notation/unit reframing; all new-physics content lives in documents 1–4.

Contents

Scope and honest typing

If gravity is the geometry-channel readout of the seam (companion note "Seam response grammar"), then horizons are not separate miracles — they all read the *same* boundary normaliser $c_3 = 1/(8\pi)$. This note collects the readouts. Markers: [P] standard horizon physics rewritten in TFPT/seam units (no new claim), [I] an exact compiler connection (audit-level), [N] a numerical readout (reproduced), [A] a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

Unit conventions used throughout

Three unit systems appear; the seam factor $c_3 = 1/(8\pi)$ is the same number in all of them, only the carried dimensionful constants change:

system	convention	Hawking temperature
Planck	$G{=}\hbar{=}c{=}k_B{=}1$	$T_H = c_3/M = 1/(8\pi M)$
reduced Planck	$\bar{M}_{\rm Pl} = M_{\rm Pl}/\sqrt{8\pi}$, $\hbar{=}c{=}k_B{=}1$	$T_H = c_3\,\bar{M}_{\rm Pl}^2/M$
SI	restore $G, \hbar, c, k_B$	$T_H = \dfrac{\hbar c^3}{8\pi G k_B M} = c_3\,\dfrac{\hbar c^3}{G k_B M}$

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

Standard references for the formulas reframed here: Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling $t_{\rm scr} \sim \beta \log S$ (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit c_3.

1 The seam constant is the universal horizon temperature factor

Every Killing horizon has $T = \hbar\kappa/(2\pi c k_B)$. Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \qquad \frac{1}{8\pi} = c_3} \qquad \text{[I]},$$

the universal factor *is* the seam constant: $T_{\rm hor} = 4c_3\,\hbar\kappa/(ck_B)$. Black holes, de Sitter and Unruh therefore share *one* thermal grammar. [P] [`verification/v8_horizon.py`]

2 Schwarzschild thermodynamics in four c_3-lines

For a non-rotating, uncharged hole in Planck units ($G{=}\hbar{=}c{=}k_B{=}1$):

$$\boxed{T_H = \frac{c_3}{M}, \qquad S_{BH} = \frac{M^2}{2c_3}, \qquad P_H = \frac{c_3}{1920\,M^2}, \qquad \tau_{\rm evap} = \frac{640}{c_3}M^3} \qquad \text{[P]}.$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is $c_3 = 1/(8\pi)$ — the same P1 boundary normaliser; [I]
- the radiated-power denominator is $1920 = |W(D_5)|$, the Weyl-group order of the D_5 carrier. [I]

Scope of P_H: this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the c_3-form is the convention statement, not a general phenomenological prediction.

object	T_H	**lifetime / entropy**
solar-mass hole	6.17×10^{-8} K	$\tau_{\rm evap} \approx 2.1 \times 10^{67}$ yr; $S/k_B \approx 1.05 \times 10^{77}$
PBH evaporating now (1.7×10^{11} kg)	$k_B T \approx 61$ MeV	relevant for γ/ν PBH signatures

3 Page time and scrambling

Since $S_{BH} \propto M^2$ and $\tau \propto M^3$, the Page point $(M \to M/\sqrt{2})$ is

$$t_{\rm Page} = \Big(1 - \tfrac{1}{2\sqrt{2}}\Big)\tau_{\rm evap} \approx 0.6464\, \tau_{\rm evap} \qquad \text{[P]}.$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate, $I(A{:}C|B)_n \sim \lambda_2^n = (2/3)^{6n}$. **This is the same operator that builds the Standard Model:** the boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ and its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ is exactly the flavor-gap eigenvalue of Paper 2 ($\Delta_{\rm gap} = -\log(2/3)^6 = 6\log\frac{3}{2}$). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. [I] [`verification/v54_seam_horizon_keystones.py`] The scrambling prefactor carries a μ_4 signature: with $\beta = 1/T_H = 8\pi M$, $\beta/2\pi = 4M$, so

$$\boxed{t_{\rm scr} \sim |\mu_4|\, M \log S} \qquad (|\mu_4| = 4) \qquad \text{[I]-audit.}$$

4 De Sitter and the Nariai bound from the Λ closure

With the action-grammar Hubble/Λ readouts ($H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$, $\rho_\Lambda/\bar{M}_{\rm Pl}^4 \sim e^{-2\alpha_\star^{-1}}$), the asymptotic de Sitter horizon gives

$$\boxed{\frac{T_{dS}}{\bar{M}_{\rm Pl}} = 16 c_3^2\, e^{-\alpha_\star^{-1}}, \qquad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128 c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122}} \qquad \text{[P]}.$$

This is the area-law horizon capacity of the universe from $(\alpha_\star^{-1}, c_3, H_\Lambda)$ alone. The maximal black hole in de Sitter (Nariai) is

$$M_N \approx 2.16 \times 10^{22}\, M_\odot \approx 4.3 \times 10^{52}\,\text{kg}, \qquad S_N = \tfrac{1}{3} S_{dS} \qquad \text{[P]},$$

the absolute gravitational-capacity limit, not an observed structure.

5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. [I] [`verification/v101_horizon_anchor.py`]

Nariai = the anchor; the Koide 2/3 is the entropy bound [I] ([verification/v101_horizon_anchor.py])

With $f(r) = 1 - 2M/r - \Lambda r^2/3$, the horizon cubic at the maximal mass $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\rm fam}\sqrt{\Lambda})$ is, in units $1/\sqrt{\Lambda}$,

$$\boxed{t^3 - 3t + 2 = (t-1)^2(t+2)} \qquad \text{roots } (1, 1, -2),$$

with coefficients $(-3, 2) = (-N_{\rm fam}, |\mathbb{Z}_2|)$ — and the root pattern is exactly the **traceless projection of the anchor**: $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1,1,-2)$ for $a = (1,1,2)$ (compare $\chi_a = (t-1)^2(t-2)$, Paper 1). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries $S_{dS}/3 = S_{dS}/N_{\rm fam}$ and the total is $\boxed{S_{\rm Nariai}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\rm fam}}$ (deficit exactly $S_{dS}/3$). The interpolation in $x = r_b/r_c$ is the closed form $S_{\rm tot}/S_{dS} = (x^2+1)/(x^2+x+1)$ — denominator $= \Phi_3(x)$, the $N_{\rm fam}$ cyclotomic — monotone from 1 to 2/3 with its minimum exactly at the merge (Fig. 2a).
- **Three-sheet conservation.** The cubic is traceless with fixed e_2, so $\pi \sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2|\, S_{dS}$ for *every* M: the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.
- **Koide-form quotient of the roots.** $Q_{\rm geom} = \sum r_i^2/(\sum |r_i|)^2$ runs monotonically over $[\frac{3}{8}, \frac{1}{2}]$: pure dS gives $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$ (the half-step), Nariai gives $\frac{3}{8} = p_2(a)/e_1(a)^2$ — exactly the two nonzero $SU(4)_1$ conformal weights (Fig. 2b).
- **The mass line is a double cover.** $\mathrm{disc}_r \propto (1-3m)(1+3m)$ in $m = M\sqrt{\Lambda}$: branch points $m = \pm 1/N_{\rm fam}$, separation $2/3 = |\mathbb{Z}_2|/N_{\rm fam}$, deck involution = the black-hole $\leftrightarrow$ cosmological horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$ and of the split $N_{\rm fam}$-linear cover $(3x+2)(3x+5)$ of Paper 2 (Fig. 1).
- **Temperature lemma / orientation.** $|\kappa_b/\kappa_c| = (2x+1)/(x(x+2)) > 1$ for $x < 1$ (equality iff $x = 1$): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, Paper 4). *Audit reading.*

Completing the seam-unit table: $dM = c_3\,\kappa\, dA$ (first law), $M = 2c_3\,\kappa A$ (Smarr), $S \le ER/(4c_3)$ (Bekenstein), $P_H = c_3/(1920\, M^2)$, lifetime $\tau = 5120\pi M^3 = 128\, g_{\rm car} M^3/c_3$ ($128 = 2^7$ the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr $A_{\rm ext} = M^2/c_3$ with $A_{\rm ext}/A_{\rm Schw} = 1/2 = 1/|\mathbb{Z}_2|$ and $S_{\rm ext} = M^2/(4c_3)$, and the Bekenstein–Mukhanov–Hod area quantum $4\ln 3 = \ln 81 = \ln(N_{\rm fam}^4) = \ln(\text{disc of the flavor cover})$ ([P] as the QNM reading). *Honest scope:* the GR identities are textbook facts; the [I] content is six independent landings on already-load-bearing atoms with zero free parameters. The "carrier in the bulk" reading (two glue chiralities = two horizon sheets, ramification = Nariai) stays [P]. No new data test (M_N today $\sim 10^{22} M_\odot$): a structure surface, not a measurement surface.

The dual anchor: the inverse flavor response *is* the Nariai root [I] ([verification/v134_dual_anchor.py])

The Nariai pattern is not only the traceless anchor — it is stored *inside the flavor compiler* as a dual invariant. For the residue/winding operators R, L of Paper 2 and the anchor

$a = (1, 1, 2)$,

$$\boxed{d := a^\top R^{-1} = a^\top L^{-1} = (-\tfrac{1}{2}, -\tfrac{1}{2}, 1)}, \qquad d \cdot \mathbf{1} = 0, \quad d \cdot a = 1, \quad (1, 1, -2) = -2d,$$

i.e. $d = \frac{3}{2}(a - \frac{4}{3}\mathbf{1})$ is the *normalised traceless anchor*, proportional to the Nariai double-root pattern above. The invariance is exact and structural (Sherman–Morrison): with $L = R + 6\,\mathbf{1}e_1^\top$, a covector is winding-invariant iff it annihilates $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ — the winding deformation lives in the democratic direction $\mathbf{1}$, which d kills. Controls: $\mathbf{1}$, e_1 and the torsion normal $n = (5, -9, 6)$ do *not* lie on the invariance plane $(\frac{1}{4}, \frac{1}{4}, -\frac{5}{2})$, so the anchor's membership is special. Together (d, n) form the dual normal pair of the flavor boundary: d reads the traceless horizon structure, n reads first-generation torsion ($n^\top R = (8, 0, 0)$, $n^\top L = (20, 0, 0)$). *Typing:* all identities [I]; the reading — a third, purely *algebraic* leg of the flavor $\leftrightarrow$ horizon bridge, beside the shared clock spectrum ([verification/v126_clock_wall_bridge.py]) and the entropy power law ([verification/v129_entropy_power_law.py]) — stays [P].

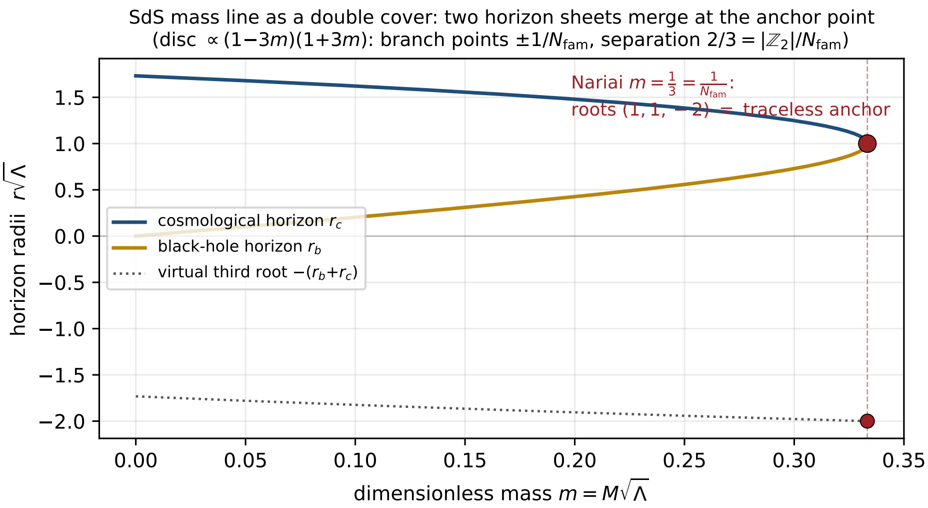

Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Nariai point $m = 1/N_{\mathrm{fam}}$, where the three roots form the traceless anchor pattern $(1, 1, -2)$. [verification/v101_horizon_anchor.py]

One orientation: the anchor is the *stationary repeller* in both sectors [I]/[P] ([verification/v102_seam_orientation.py])

The evaporation direction of the temperature lemma upgrades to exact variational structure on both one-dimensional moduli lines (Fig. 4). **Flavor:** the canonical relaxation $\dot{q} = (\Delta/N_{\mathrm{fam}})(q-2)(q-5)$ is the *gradient flow* of the cubic potential

$$V(q) = -\frac{\Delta}{N_{\mathrm{fam}}}\Big(\frac{q^3}{3} - \frac{7}{2}q^2 + 10q\Big), \qquad \boxed{V''(2) = +\Delta, \quad V''(5) = -\Delta}$$

— the critical points are exactly the two branch points, the stationary curvatures are $\pm$the transfer gap, the inflection sits at $q = \frac{7}{2}$ = scalaron/2, and the Lyapunov rate is constant: $d(-\ln\rho)/dt = \Delta$ exactly. **Gravity:** the SdS entropy functional has

$$\frac{d}{dx}\frac{S_{\mathrm{tot}}}{S_{dS}} = \frac{(x-1)(x+1)}{\Phi_3(x)^2}, \qquad \boxed{\Big(\frac{S_{\mathrm{tot}}}{S_{dS}}\Big)''(1) = \frac{2}{9} = \frac{|\mathbb{Z}_2|}{N_{\mathrm{fam}}^2} = \frac{2}{3} \cdot \frac{1}{3}}$$

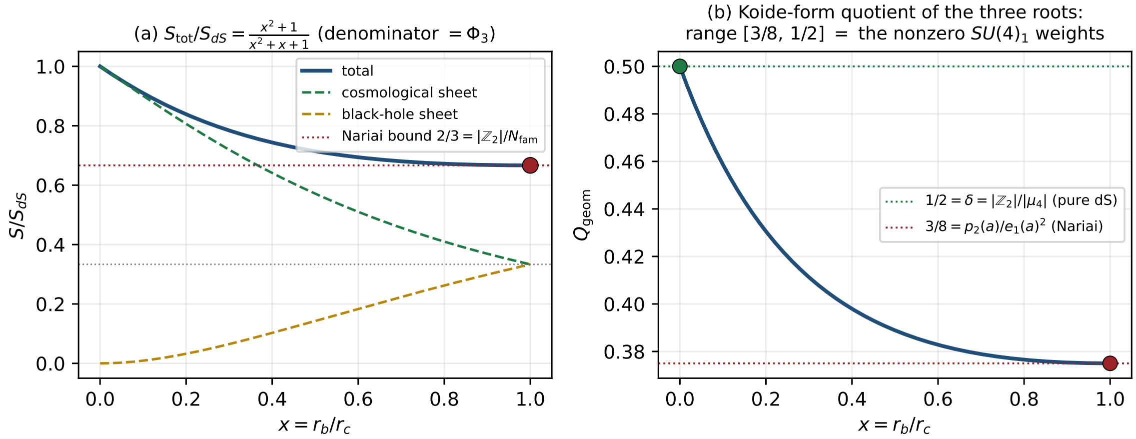

Figure 2: (a) Total SdS entropy interpolates from S_{dS} to the Nariai bound $\frac{2}{3}S_{dS}$ via $(x^2+1)/\Phi_3(x)$. (b) The Koide-form quotient of the three roots runs exactly over the nonzero $SU(4)_1$ weights $[\frac{3}{8}, \frac{1}{2}]$. [`verification/v101_horizon_anchor.py`]

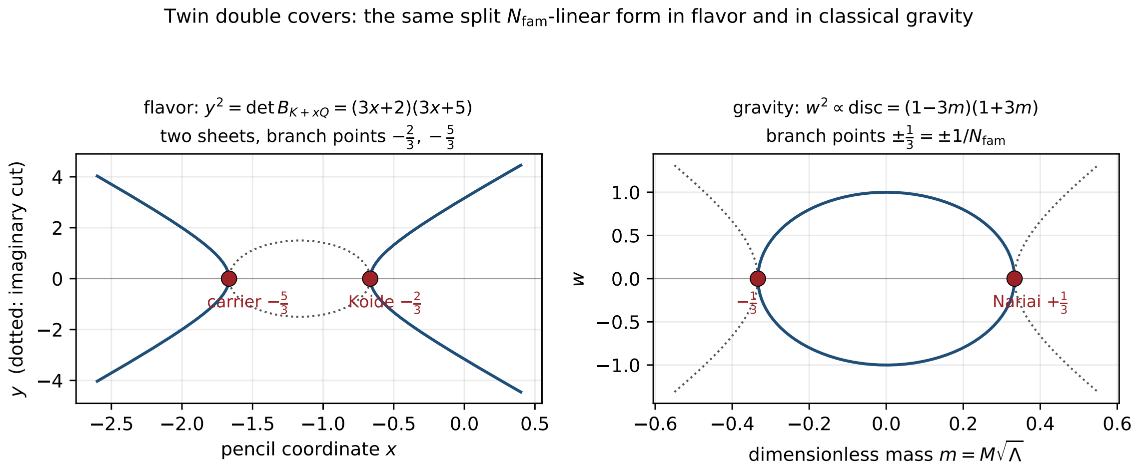

Figure 3: Twin double covers: the flavor anchor-block cover $y^2 = (3x+2)(3x+5)$ (Paper 2) and the SdS mass-line cover $w^2 \propto (1-3m)(1+3m)$ share the split N_{fam}-linear form with rational branch points at spine-over-family values. [`verification/v101_horizon_anchor.py`]

— the Nariai/anchor point is the *unique* physical stationary point, its curvature is a grammar constant (= the product of the two Nariai entropy fractions), and the temperature lemma makes the evaporation an entropy *ascent* away from it. **Orientation theorem [I]:** in both sectors the anchor-type configuration is a stationary point of the natural functional and the dynamics flows away from it, with grammar-constant stationary curvatures. *Honest disanalogies (recorded):* the flavor attractor is itself a branch point while the gravity attractor is the smooth endpoint; the flavor functional is the potential of a [P]-typed relaxation while the gravity functional is the Bekenstein–Hawking entropy — "one variational principle of the seam" stays a *reading* [P], and no open gate is touched.

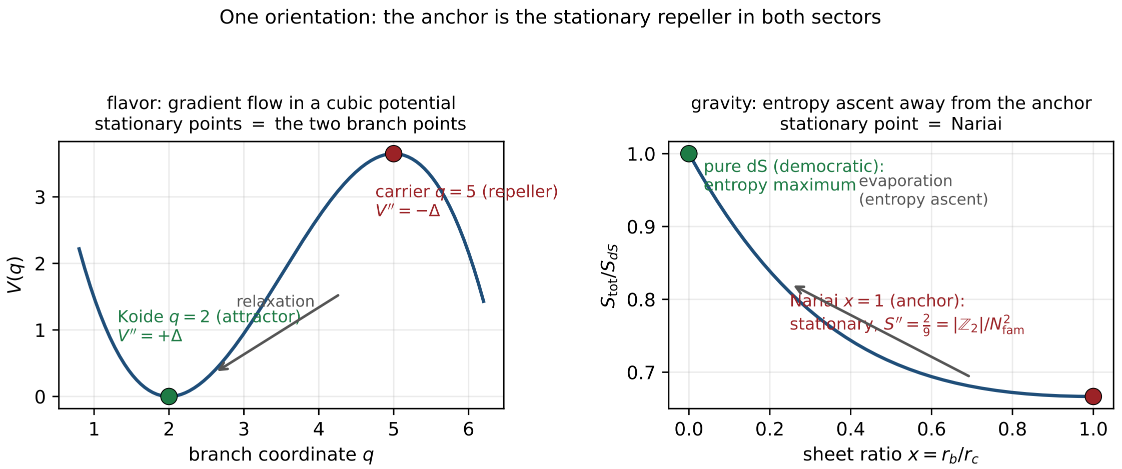

Figure 4: The orientation theorem: the anchor configuration is the stationary repeller of the natural functional in both sectors — flavor (cubic potential, curvatures $\pm\Delta$) and gravity (SdS entropy, curvature $2/9 = |\mathbb{Z}_2|/N_{\mathrm{fam}}^2$). [`verification/v102_seam_orientation.py`]

The trisection normal form: the canonical coordinate exists [I] ([`verification/v103_trisection_normal_form.py`])

The curvature-matching question of the previous box has a closed answer. The SdS horizon cubic is uniformized by **angle trisection**: with $r = 2\cos\theta$ (units $1/\sqrt{\Lambda}$), $r^3 - 3r = 2\cos 3\theta$ *exactly*, so the cubic is $\cos 3\theta = -3m$ — the three horizon roots are the three trisection branches (the $\mathbb{Z}_3$ rotation = the triality deck, cf. coker $Q = \mathbb{Z}/N_{\mathrm{fam}}$). In the centered angle ψ ($\cos\varphi = -3m$, $\psi = \varphi - \pi$; Nariai $\psi = 0$, pure dS $\psi = \pi/2$, deck $\psi \to -\psi$):

$$m = \frac{\cos\psi}{N_{\mathrm{fam}}}, \qquad \boxed{\frac{S_{\mathrm{tot}}}{S_{dS}} = \frac{4}{3} - \frac{2}{3}\cos\frac{2\psi}{3}}$$

— *one* cosine whose mean ($|\mu_4|/N_{\mathrm{fam}}$), amplitude and frequency (both $|\mathbb{Z}_2|/N_{\mathrm{fam}}$) are glue atoms over the family count (Fig. 5). Three exact consequences:

- **Canonical curvature = the Koide constant to the family power:** $\sigma''(\psi{=}0) = \frac{8}{27} = (\frac{2}{3})^3 = (|\mathbb{Z}_2|/N_{\mathrm{fam}})^{N_{\mathrm{fam}}}$.
- **Invariant base slope:** $d\sigma/dm|_{\mathrm{Nariai}} = -\frac{8}{9} = -\operatorname{rank} E_8/N_{\mathrm{fam}}^2$ (coordinate-invariant, cross-checked in both coordinates; generally $-\frac{4}{3}\sin(2\psi/3)/\sin\psi$); dimensionful $dS_{\mathrm{tot}}/dM|_N = -r_N/(N_{\mathrm{fam}}c_3)$.
- **The bridge:** the flavor invariant is a *rate*, multiplier $(2/3)^6 = (2/3)^{2N_{\mathrm{fam}}}$ per transport step; the gravity invariant is a *curvature*, $(2/3)^{N_{\mathrm{fam}}}$ in the canonical angle — same base $2/3 = |\mathbb{Z}_2|/N_{\mathrm{fam}}$, exponent ratio $2 = |\mathbb{Z}_2|$. What is still missing for a full identification is the gravity-side *clock* (the near-Nariai evaporation generator that converts curvature into a rate) — the same class of missing object as the Koide transfer generator of Paper 4: **one clock, two known geometries**. That identification stays [P].

The classical clock speaks anchor — and the honest (2/3)-test [I]/[P] ([`verification/v104_nariai_clock.py`])

The clock question of the previous box has a *classical* half, and it is pure GR (Ginsparg–Perry 1983; Bousso–Hawking class): linearize around the Nariai geometry $dS_2 \times S^2$. **Static**

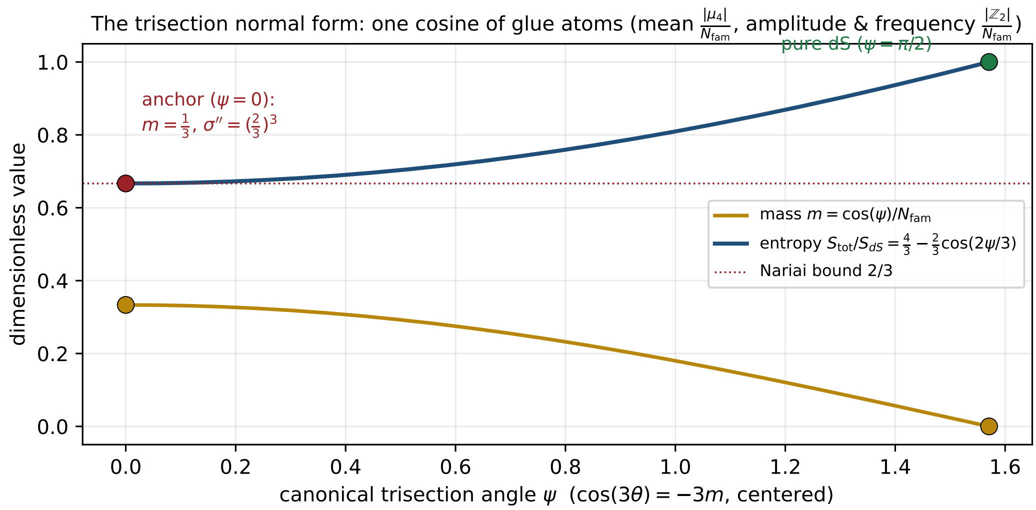

Figure 5: The trisection normal form: in the canonical angle ψ the mass and the total entropy are pure (co)sines built from glue atoms; the anchor sits at $\psi = 0$ with canonical curvature $(2/3)^3$. [`verification/v103_trisection_normal_form.py`]

pin [I]: on the dS_2 static patch the function $\phi(\rho) = \rho$ satisfies $\frac{d}{d\rho}[(1 - \Lambda\rho^2)\phi'] = -2\Lambda\phi$ *exactly* — and this static mode *is* the exact SdS two-horizon deformation, so the S^2-radius modulus mass is pinned internally: $m^2 = -2\Lambda = -|\mathbb{Z}_2|\Lambda$ (the Laplace-type form of the linearization is the one external GR import, typed honestly). The Euclidean spectrum is the Ginsparg–Perry tower $(l(l+1) - 2)\Lambda$: exactly *one* negative mode $-|\mathbb{Z}_2|\Lambda$. **The clock [I]:** in Hubble units the homogeneous modes $\phi \sim e^{\lambda H t}$ obey

$$\boxed{\chi_{\mathrm{clock}}(\lambda) = \lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2)}$$

— **the anchor quadratic**: eigenvalues $\{1, -2\}$ = the distinct anchor roots, and the Nariai cubic factors as $(t-1)^2(t+2) = (t-1)\cdot\chi_{\mathrm{clock}}(t)$. The anchor thus appears a *third* time: as configuration roots, as the curvature base, and as the spectrum of the linearized clock. The horizon split is linear in the canonical angle with coefficient $1/\sqrt{N_{\mathrm{fam}}}$, and the entropy deviation grows at exactly $2H = |\mathbb{Z}_2|\times$Hubble. **Honest (2/3)-test [P]:** the classical clock eigenvalues are *integers*, not (2/3)-powers — the (2/3)-powers stay in the functional (curvature $(2/3)^{N_{\mathrm{fam}}}$). The conversion curvature→rate at one loop (quantum (anti-)evaporation) remains external QFT input: the bridge to the flavor rate $(2/3)^{2N_{\mathrm{fam}}}$ is *not* closed here. The seam variational principle now has exactly *one* missing object: the quantum clock.

The quantum-clock target, made quantitative [I]/[P] ([`verification/v107_quantum_clock_target.py`])

Instead of guessing the quantum clock, its job description can be pinned exactly. **(1) The classical clock already lives on the cusp ladder [I]:** across the physical Ginsparg–Perry modes ($l = 0$: $(\lambda-1)(\lambda+2)$; $l = 1$: $\lambda(\lambda+1)$) the decay exponents are

$$\boxed{\{0, -1, -2\} = -\,\mathrm{Spec}(V) = -N_{\mathrm{fam}} \times \text{cusp weights}}$$

— exactly the grading that carries the transfer operator's eigenspaces (Papers 2/6); the $l \geq 2$ modes are damped oscillations with $\mathrm{Re}\,\lambda = -\frac{1}{2} = -\delta$ (audit). **(2) The exact target [I]:** the transfer rates $\{0, \Delta, 6\ln 3\}$ are *not* weight-linear — $\mathrm{rate}(2)/\mathrm{rate}(1) = \log_{3/2} 3 = 1 + \log_{3/2} 2 = 2.7095$, with the exact identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$: *the quantum clock must bend the weight-linear spectrum by* $\log_{9/4} 3 = 1.35476$ *per weight step.* **(3) Coupling landscape [I]:** the dimensionless clock coupling $\kappa = (c/24\pi)(\Lambda/\bar{M}_{\mathrm{Pl}}^2)$ with $c = 8$ is 7.6×10^{-122} at the cosmological Nariai (hopeless) and $\kappa = 1/(3\pi) = 0.106$ at seam curvature — the only regime where an $O(0.35)$ bend is reachable, and it is borderline non-perturbative: one-loop linear response cannot fix the bend; an $O(1)$ resummation or an exact seam construction is required. *Status [P]:* R1 is sharpened from "find a clock" to "produce the bend $\log_{3/2} 3$ on the cusp ladder at coupling $O(1/3\pi)$". Nothing is fitted; the identification stays open.

The resummed clock: the bend has a closed form [I]/[P] ([verification/v124_resummed_clock.py])

The "bend" is not an exotic target — the frozen transfer spectrum *is* a resummed logarithm. **(1) Spectrum reading [I]:** the eigenvalues $\{1, (\frac{2}{3})^6, (\frac{1}{3})^6\}$ are exactly

$$\boxed{\lambda_n = \Big(1 - \frac{n}{N_{\mathrm{fam}}}\Big)^{p_2}, \qquad n = 0, 1, 2} \qquad (p_2 = 6 = |R^+(A_3)|),$$

so the clock has the closed form $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\mathrm{fam}})$ — it reproduces $\{0, \Delta, 6\ln 3\}$, and the v107 bend $\log_{3/2} 3$ becomes a one-line consequence. **(2) The linearisation carries the sheet [I]:** $\mathrm{rate}(n) = |\mathbb{Z}_2|\, n + \frac{n^2}{3} + \frac{2n^3}{27} + \cdots$ — the one-loop term has slope exactly $|\mathbb{Z}_2| = 2$ relative to the *integer* classical Ginsparg–Perry clock $\{0, -1, -2\}$: the quantum clock is the sheet-doubled classical clock plus the geometric tail $\sum_k (n/N)^k/k$. **(3) The wall at N_{fam} [I]:** rate $\to \infty$ as $n \to N_{\mathrm{fam}}$ — the resummed logarithm has a pole at the family count, so the cusp ladder *must* end at $n = N_{\mathrm{fam}} - 1 = 2$: the three-level transfer spectrum is forced by the clock itself. *R1 re-sharpened [P]:* the semiclassical job is now "derive $-p_2 \ln(1 - n/N_{\mathrm{fam}})$ from the near-Nariai one-loop at $\kappa = 1/(3\pi)$" — a closed-form target, with the linear-response slope $|\mathbb{Z}_2|$ as its first checkpoint. All spectra frozen (v54/v69/v95/v104/v107); nothing fitted.

The clock and the wall share a spectrum [I]/[P] ([verification/v126_clock_wall_bridge.py])

Both ends of the resummed clock connect to established exact objects. **(1) The weights are the parabolic weights [I]:** the arguments $n/N_{\mathrm{fam}} = \{0, \frac{1}{3}, \frac{2}{3}\}$ of the resummed logarithm are *exactly* $\mathrm{spec}\, A_0^\star$ — the spectrum of the exact (U_{wall}) residue (research contracts):

$$\boxed{\lambda_n = (1 - \alpha_n)^{p_2}, \qquad \alpha_n \in \mathrm{spec}\, A_0^\star = \{0, \tfrac{1}{3}, \tfrac{2}{3}\}}$$

— the horizon clock and the flavor wall share their spectrum through the complement map $\alpha \mapsto 1 - \alpha$. **(2) Checkpoint 1 passes classically [I]:** the clock's linear coefficient $p_2/N_{\mathrm{fam}} = 2 = |\mathbb{Z}_2|$ *equals* the established classical entropy-deviation rate $2H = |\mathbb{Z}_2| \times \mathrm{Hubble}$ (v104) — the slope-2 linear response is already the classical entropy rate; the genuinely *quantum* content of R1 isolates to the geometric tail $\sum_{k \geq 2} \alpha^k/k$. **(3) The determinant is the entropy curvature [I]:** $\det(\mathbf{1} - A_0^\star) = \frac{2}{9} = |\mathbb{Z}_2|/N_{\mathrm{fam}}^2$ — exactly the gravity entropy curvature at the anchor (v102), a fourth independent appearance; $\mathrm{tr}(\mathbf{1} - A_0^\star) = 2 = |\mathbb{Z}_2|$, $\det A_0^\star = 0$. *R1 re-targeted [P]:* derive "rate $= -p_2 \ln(1 - \alpha)$" with α the Mehta–Seshadri parabolic weight — first order classical, tail quantum: **one spectrum, two gates, one object $A_0^\star$.**

The clock is a log-determinant: the tail is the ring (RPA) resummation [I]/[P] ([verification/v127_ring_resummation.py])

The "$O(1)$ resummation" demanded by v107 is now *typed.* For a rank-one insertion P ($\operatorname{tr} P^k = 1$), $-\ln\det(\mathbf{1} - \alpha P) = -\ln(1-\alpha) = \sum_k \alpha^k/k$, so the resummed clock is exactly

$$\boxed{\text{rate} = -p_2 \operatorname{Tr}\ln(\mathbf{1} - \alpha P)}$$

— $p_2 = 6$ *identical* one-loop ring (RPA) towers, **one per hexagon site** (= positive root of A_3), with coupling α = the parabolic weight. *Ring ladder [I]:* the k=1 ring is the *classical* term ($6\alpha = |\mathbb{Z}_2|n$, the v104/v126 entropy rate); the k=2 ring is the first quantum correction ($\frac{1}{3}$ at weight 1); the partial sums converge to $\{\Delta, 6\ln 3\}$ monotonically from below. *Coupling cross-link [I] (audit):* the established seam coupling is $\kappa = \frac{c}{24\pi} = \frac{8}{24\pi} = \frac{1}{3\pi} = \alpha_1/\pi$ — the *first parabolic weight over* π. *The wall [I]:* the ring series diverges at $\alpha \to 1$ ($n \to N_{\text{fam}}$) — the v124 wall is the standard RPA instability. *R1 final form [P]:* show that the near-Nariai one-loop effective action of the seam mode, with insertion = parabolic weight and multiplicity p_2, is this ring sum — "**the seam clock is the RPA determinant of one mode on six sites**". The resummation type is identified; the derivation itself remains open.

Where the six sites live ([verification/v128_graded_hull.py]): the Ginsparg–Perry $l = 1$ sector — the *zero modes* $\lambda \in \{0, -1\}$ of the classical clock (v104/v107) — has $2l + 1 = 3$ modes, and sheet doubling gives $2 \times 3 = 6 = p_2$ [I]: the ring multiplicity equals the sheet-doubled zero-mode count. [P] reading (recorded): zero modes are exactly the modes that force nonperturbative resummation in de Sitter one-loop physics — the natural home of the six RPA towers.

The clock is an entropy power law [I]/[P] ([verification/v129_entropy_power_law.py])

What the resummation argument $1 - n/N_{\text{fam}}$ *is* physically: the three established Nariai entropy levels (v101) are $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (pure dS; two-horizon total; single horizon — deficit $S_{dS}/3$ per step) — exactly the complementary cusp weights. Hence [I]:

$$\boxed{\lambda_n = \left(\frac{S_n}{S_{dS}}\right)^{p_2}, \qquad \text{rate}(n) = p_2 \ln\frac{S_{dS}}{S_n}}$$

— the transfer eigenvalues are the *entropy fractions to the hexagon power*; the rates are hexagon-weighted log-entropy deficits. *The coupling is the entropy deficit [I] (triple identity):* $\alpha = n/N_{\text{fam}}$ = the Mehta–Seshadri parabolic weight (v115/v126) $= 1 - S_n/S_{dS} = \Delta S_n/S_{dS}$ — the RPA insertion strength is the fractional entropy deficit; the coupling is not a free parameter in *any* of its three readings. *Orientation [I]:* weight 0 $\leftrightarrow$ pure dS (maximal entropy, stationary attractor), weight 2 $\leftrightarrow$ single horizon (fastest decay) — the established repeller$\to$attractor flow. *R1, thermodynamic form [P]:* derive $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ from the near-Nariai one-loop — a Gibbons–Hawking-type entropy power law with exponent = the hexagon size (audit: $p_2 = 6 = 2h$ reads as the two-point form of a weight $h = 3 = N_{\text{fam}}$ operator; recorded, not claimed).

The Born square ([verification/v130_born_square.py]): the weight audit is now *derived* from mode counting [I]. In the standard collective-coordinate treatment each zero mode contributes a moduli factor $\propto S^{1/2}$, so with $n_{\text{zero}} = 6$ the transition *amplitude* scales as $A \sim (S/S_{dS})^3$ and the *probability* is its Born square $\Gamma = |A|^2 = (S/S_{dS})^6$:

$$\boxed{h = \tfrac{n_{\text{zero}}}{2} = 3 = N_{\text{fam}}, \qquad p_2 = 2h \text{ (the Born rule)}}$$

— and the 3 is the dimension of the l=1 eigenspace = the number of *proper conformal Killing vectors of* S^2 (the classic near-Nariai zero modes, Ginsparg–Perry/Bousso–Hawking class), while the $\times 2$ is the *horizon pair* = the SdS mass-line deck (v101). The same 3 counts the generations and the sphere's conformal directions. *Residue [P]:* justify the per-zero-mode $S^{1/2}$ normalisation on the Nariai instanton for exactly these six modes — a textbook-class computation; no exotic object remains in R1.

The measure is the area ([verification/v131_measure_is_area.py]): the geometric core of that normalisation is exact [I]: on $S^2(r)$ the L^2 norm of *each* of the three normalised $l{=}1$ harmonics is

$$\boxed{\|Y_{1m}\|^2 = r^2 = \frac{A}{4\pi}}$$

(all three integrated exactly) — the zero-mode norm *is* the horizon area, hence the entropy. The collective-coordinate Jacobian per mode is the mode norm $\sim S^{1/2}$ (standard change of variables), so: norm = area $\to$ Jacobian = $S^{1/2}$ $\to$ six modes $\to$ amplitude S^3 $\to$ Born $\to$ S^6 — every exponent factor carries a derivation-level origin. The 4π is the same Gauss–Bonnet primitive that builds $c_3 = \frac{1}{2}(4\pi)^{-1}$, and it *cancels* in all ratios — exactly why the clock sees only entropy fractions. *The last step [P]:* show $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ (non-zero-mode determinant cancellation; same local near-Nariai geometry) — the one remaining computation of R1.

The det-ratio anomaly is the Koide constant [I]/[P] ([verification/v132_detratio_anomaly.py])

The det-ratio question acquires its first *exact* invariant. The Ginsparg–Perry sphere operator on the unit S^2 is $L = -\Delta - 2$ (the $-2 = -|\mathbb{Z}_2|$ is the v104 modulus mass), with eigenvalues $(l{+}2)(l{-}1)$ and kernel the three $l{=}1$ zero modes. Its heat coefficient is made of atoms [I]: $a_1 = \frac{R}{6} - E = \frac{1}{3} + 2 = \frac{1}{N_{\text{fam}}} + |\mathbb{Z}_2| = \frac{7}{3}$, and the scaling anomaly of the non-zero-mode determinant is

$$\boxed{\zeta(0)\big|_{\det'} = a_1 - \dim\ker = \tfrac{7}{3} - 3 = -\tfrac{2}{3} = -\frac{|\mathbb{Z}_2|}{N_{\text{fam}}}}$$

— confirmed independently by numerical continuation of the heat trace ($\sim 10^{-7}$): **the det-ratio anomaly is the Koide constant**, the eighth appearance of $2/3$ and the first as a *spectral* anomaly. *Consequence [I]:* $\det'(cL) = c^{\zeta(0)}\det'(L)$ — the determinant ratio between configurations is *not* automatically 1; the S^2-sector share of the scaling budget is exactly $-\frac{2}{3}$ per Hubble-unit conversion, while the dimensionless tower itself is configuration-independent (pure-number GP spectrum). *The budget statement [P] (the remaining step, made precise):* show the *full* $\zeta(0)$ budget of the 4d Nariai fluctuation problem (dS_2 + gauge/ghost + this S^2 sector) vanishes or is absorbed into the established power law — the S^2 share is pinned exactly; nothing is claimed about the total.

The zeta budget: the reduced route carries the anchor ratios [I]/[P] ([verification/v133_zeta_budget.py])

Euclidean Nariai is $S^2 \times S^2$ — the temporal sector continues to the *same* unit sphere, so both candidate determinant routes are exactly computable. *(i) The reduced route [I]:* each 2d sector carries the v132 operator with $\zeta(0)|_{\det'} = -\frac{2}{3}$; the two sectors give

$$\boxed{\text{budget}_{\text{red}} = -\tfrac{2}{3} - \tfrac{2}{3} = -\tfrac{4}{3} = -\frac{e_1}{p_0} \text{ (minus the seed gain)}}$$

— the per-sector share and the total are *two of the three v119 anchor-ratio triad values*; the six zero modes split $3+3$ across the two spheres: the sheet doubling *is* the two-sphere structure of Euclidean Nariai (= the horizon pair, v101). *(ii) The naive 4d route [I]:* with the standard S^2 heat expansion $1/t + \frac{1}{3} + t/15$ and the symmetric split, $a_2(4d) = (\frac{4}{3})^2 + 2 \cdot \frac{9}{10} = \frac{161}{45}$, six zero modes, $\zeta(0)|_{\det',4d} = \frac{161}{45} - 6 = -\frac{109}{45}$ — *no* anchor-atom reading. *(iii) Route discrimination [P]:* the budget computation structurally favours the **reduced seam reading** (pure anchor-ratio anomalies), consistent with the theory's 2d-seam architecture throughout — route-selection evidence, not proof. *Remaining [P]:* the graviton/ghost shares on $S^2 \times S^2$ (Volkov–Wipf class, external standard) — a finite list of known heat coefficients.

The Volkov–Wipf firewall: the external edge is two copies of the reduced budget [I]/[P] ([verification/v138_vw_firewall.py])

The "finite list" above is now *filled in externally.* For the full one-loop graviton path integral on $S^2 \times S^2$ the log-coefficient is (arXiv:2506.02142, Table 4.1, agreeing with Volkov–Wipf, Nucl. Phys. B582 (2000) 313):

$$\alpha_{\text{bulk}} = \frac{22}{45}, \qquad \alpha_{\text{edge}} = -\frac{8}{3}, \qquad \alpha_{\text{PI}} = -\frac{98}{45}$$

(no anchor atom; the difference to the reduced budget is $-\frac{38}{45}$ with $38 = 2 \cdot 19$). The structural finding: the external *edge* factor consists of **two identical copies** (one per horizon), and

$$\boxed{\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times \left(-\frac{4}{3}\right)}$$

— the per-copy magnitude $\frac{4}{3} = e_1/p_0$ is *exactly* the reduced seam budget above (the seed gain; $-\frac{2}{3}$ per sector). The TFPT reduced route lands on the **edge sector** of the external bulk/edge split; the bulk ($\frac{22}{45}$) is the ideal graviton gas, which the seam clock does not contain; two copies = the horizon pair = the sheet doubling (v101; the 3+3 zero-mode split). *Typing decision [P] (recorded):* R1 = the reduced zero-mode/RPA seam clock = an edge-sector object; the full 4d Volkov–Wipf determinant is the *gravity firewall* (its $-\frac{98}{45}$ is absorbed in the $\mu_0/(\Lambda G)^3$ normalisation), not a TFPT atom. Per-copy sign/orientation (the edge factor enters inverted) and the Lorentzian edge-mode reading stay [P].

6 Turnaround radius, gravitational waves, light

Bound structures. In a Λ-universe the maximal bound radius is $r_{\text{ta}}(M) = (GM[2\pi e^{\alpha_\star^{-1}}/\bar{M}_{\text{Pl}}]^2)^{1/3}$ — halo, cluster, supercluster and void scales follow from the Λ closure with no new parameter. [P]

Gravitational waves. The low-curvature branch is $R + R^2$; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron, $M_{\text{scal}} \approx 3.06 \times 10^{13}$ GeV) is frozen for astrophysical frequencies. Hence

$$\boxed{v_{\text{GW}} = c} \qquad \text{(no measurable dispersion)} \qquad \text{[P]},$$

a clean falsifier: a genuine $v_{\text{GW}} \neq c$ would break this gravity closure.

Light. TFPT does not predict the SI number c (definitional); it predicts the *universality of the causal cone* — RP → OS reconstruction → one Lorentzian cone shared by EM, gravity and massless modes ($v_\gamma = v_g = c$, no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\boxed{\beta_{\text{rad}} = \frac{u}{4\pi} \approx 0.2424^\circ} \qquad \text{[N]},$$

from the same seed $u = \phi_0^{\text{ret}}$ that fixes the Cabibbo angle, Ω_b and θ_{13}.

Black-hole mechanics in seam units ($c_3 = 1/(8\pi)$): every coefficient is a compiler atom

quantity	standard form	seam form	compiler atom
Einstein equations	$G_{\mu\nu} = 8\pi T_{\mu\nu}$	$G_{\mu\nu} = T_{\mu\nu}/c_3$	$c_3 = 1/(8\pi)$ (P1)
first law	$dM = \frac{\kappa}{8\pi} dA$	$dM = c_3 \kappa\, dA$	seam constant
Smarr (Schw.)	$M = \frac{\kappa A}{4\pi}$	$M = 2c_3 \kappa A$	
temperature	$T_H = \frac{\kappa}{2\pi}$	$T_H = 4c_3 \kappa$	$1/(2\pi) = 4c_3$
entropy	$S = \frac{A}{4}$	$S = 2\pi c_3 A$	$1/4 = 1/\lvert\mu_4\rvert$
Bekenstein bound	$S \leq 2\pi ER$	$S \leq ER/(4c_3)$	
Hawking power	$P = \frac{1}{15360\, \pi M^2}$	$P = \frac{c_3}{1920\, M^2}$	$1920 = \lvert W(D_5)\rvert$
lifetime	$\tau = 5120\, \pi M^3$	$\tau = 128\, g_{\mathrm{car}} M^3/c_3$	$128 = 2^7$, $7 =$ scalaron
Kerr extremal	$A_{\mathrm{ext}} = 8\pi M^2$	$A_{\mathrm{ext}} = M^2/c_3$	$A_{\mathrm{ext}}/A_{\mathrm{Schw}} = 1/\lvert\mathbb{Z}_2\rvert$
area quantum (Hod)	$\Delta A = 4 \ln 3$	$\Delta A = \ln(N_{\mathrm{fam}}^4)$	$81 =$ disc of the cover
Nariai bound	$S_N = \frac{2\pi}{\Lambda}$	$S_N = \frac{2}{3} S_{dS}$	$2/3 = \lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ (Koide)

Figure 6: Black-hole mechanics in seam units: every classical coefficient is a compiler atom. [`verification/v101_horizon_anchor.py`]

7 Search targets (not claims)

Audit-level search ansätze [A]

- **Black-hole echoes:** any near-horizon echo amplitude ratio would be a natural transfer-rate target, $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$. A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel $I \sim (2/3)^{6n}$ is a falsifiable shape if a boundary-recovery rate is ever measured.

Net

All horizon formulas share the one seam normalisation $1/(2\pi) = 4c_3$. Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii, $v_{\mathrm{GW}} = c$ and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$ in Hawking power, $|\mu_4| = 4$ in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

Appendix: computational verification

The seam-unit identities of this note (the thermal grammar $\frac{1}{2\pi}{=}4c_3$, the power denominator $1920{=}|W(D_5)|$, the de Sitter entropy form, the Page fraction, the scrambling $|\mu_4|{=}4$, and $\beta_{\mathrm{rad}}{=}\varphi_0^{\mathrm{ret}}/4\pi{=}0.2424°$) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same $c_3{=}1/(8\pi)$ primitive as the rest of the suite. The search-target ansätze remain [A] and are deliberately not part of the pass/fail checks.